A PROJECT TITLE

ON

Crop Yield Prediction Using Random Forest, XGBoost, and Support Vector Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

MD AMAN	2520030190
V Sree Pragnya	2520030238

Under the Esteemed guidance of

Mrs. JYOTHSNA DATTI

ASSISTANT PROFESSOR

Department of Computer Science and Engineering

Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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1.ABSTRACT:

Agriculture is one of the most important sectors as it provides food, employment, and supports the economy. Accurate crop yield prediction is important for agricultural planning, resource management, and improving farming decisions. Crop yield is influenced by several factors, including crop type, temperature, rainfall, humidity, soil characteristics, fertilizer usage, pesticide usage, and cultivated area. Traditional methods of estimating crop yield may not effectively capture the complex relationships among these factors.This project aims to develop and compare machine learning models for crop yield prediction using Random Forest, XGBoost, and Support Vector Regression (SVR). The project uses the Crop Yield Data with Soil and Weather Dataset, which contains historical agricultural records along with soil and weather-related information. The dataset includes features such as crop type, cultivated area, fertilizer usage, pesticide usage, temperature, rainfall, humidity, and soil characteristics such as nitrogen, phosphorus, potassium, and pH.The collected data is cleaned and preprocessed before training the models. Random Forest, XGBoost, and Support Vector Regression models are trained to predict crop yield. The performance of the models is evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score. The model with the best performance will be selected for crop yield prediction.The proposed system can help analyze the relationship between agricultural, soil, and weather factors and crop yield. It can support farmers and agricultural planners in making better decisions related to crop planning, fertilizer and pesticide management, and efficient utilization of agricultural resources.

2. INTRODUCTION:

Agriculture is one of the most important sectors of the economy, providing food, employment, and raw materials for various industries. Crop yield depends on several factors, including rainfall, temperature, soil quality, irrigation, fertilizer usage, and crop type. Since these factors can vary significantly across different regions and seasons, accurately predicting crop yield is challenging. Traditional methods of estimating crop production often rely on historical records and manual analysis, which may not provide sufficiently accurate or timely results. Machine learning offers an effective approach by analyzing large amounts of agricultural data and identifying patterns that can be used for reliable crop yield prediction.

This project, “Crop Yield Prediction Using Random Forest, XGBoost, and Support Vector Regression,” focuses on developing and comparing machine learning models to predict crop yield based on important agricultural and environmental parameters. The models are trained using historical crop data and evaluated using performance measures such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 Score. Random Forest, XGBoost, and Support Vector Regression are selected because they can effectively capture complex relationships between input factors and crop yield. The proposed system aims to provide accurate predictions that can help farmers and agricultural organizations make better decisions regarding crop selection, irrigation, fertilizer application, and overall farm management.

3. SOFTWARE REQUIREMENTS:

The proposed crop yield prediction system is developed using Python and machine learning libraries to build and compare different prediction models. Google Colab or Jupyter Notebook is used for coding, data preprocessing, model training, and testing. The project can be executed on a standard computer with a stable internet connection and the required software installed.

Component	**Requirement**
Operating System	Windows 10/11
Processor	Intel Core i3 or above
RAM	4 GB (8 GB Recommended)
Programming Language	Python 3.x
IDE	Jupyter Notebook / Google Colab
Libraries	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, XGBoost
Dataset	Crop Yield Prediction Dataset

DATASET: "Kaggle: Crop Yield Data with Soil and Weather Dataset".

https://www.kaggle.com/datasets/anshumish/crop-yield-data-with-soil-and-weather-dataset

Signature of the Guide **Course Instructor**